

BOIN12: Bayesian Optimal Interval Phase I/II Trial Design for Utility-Based Dose Finding in Immunotherapy and Targeted Therapies

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abstract

PURPOSE For immunotherapy, such as checkpoint inhibitors and chimeric antigen receptor T-cell therapy, where the efficacy does not necessarily increase with the dose, the maximum tolerated dose may not be the optimal dose for treating patients. For these novel therapies, the objective of dose-finding trials is to identify the optimal biologic dose (OBD) that optimizes patients' risk-benefit trade-off.

METHODS We propose a simple and flexible Bayesian optimal interval phase I/II (BOIN12) trial design to find the OBD that optimizes the risk-benefit trade-off. The BOIN12 design makes the decision of dose escalation and de-escalation by simultaneously taking account of efficacy and toxicity and adaptively allocates patients to the dose that optimizes the toxicity-efficacy trade-off. We performed simulation studies to evaluate the performance of the BOIN12 design.

RESULTS Compared with existing phase I/II dose-finding designs, the BOIN12 design is simpler to implement, has higher accuracy to identify the OBD, and allocates more patients to the OBD. One of the most appealing features of the BOIN12 design is that its adaptation rule can be pretabulated and included in the protocol. During the trial conduct, clinicians can simply look up the decision table to allocate patients to a dose without complicated computation.

CONCLUSION The BOIN12 design is simple to implement and yields desirable operating characteristics. It overcomes the computational and implementation complexity that plagues existing Bayesian phase I/II dose-finding designs and provides a useful design to optimize the dose of immunotherapy and targeted therapy. User-friendly software is freely available to facilitate the application of the BOIN12 design.

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INTRODUCTION

The advent of targeted therapy and immunotherapy, such as checkpoint inhibitors and chimeric antigen receptor T-cell therapy, has revolutionized cancer treatment. For these novel therapies, although toxicity typically increases with the dose, efficacy may plateau or even decrease at high doses.^{1,2} In addition, some of these therapies demonstrate minimal toxicity even in high doses, making the maximum tolerated dose (MTD) unlikely to be reached.³ As a result, the traditional dose-finding design paradigm, which was developed to find the MTD, may not be suitable for targeted therapy and immunotherapy.

Consider five doses of a targeted or immunotherapy agent $d = 1, 2, 3, 4$, and 5 , with true toxicity probabilities $p_T = .05, .12, .27, .35$, and $.50$. If the true efficacy probabilities are $p_E = .20, .35, .36, .37$, and $.38$, then efficacy reaches a plateau of approximately 0.35 at $d = 2$. The conventional dose-finding methods (eg, $3 + 3$ algorithm or continuous reassessment

method [CRM] with a target toxicity probability of $.3$) are most likely to select $d = 3$ as the MTD and use it for cohort expansion (CE) or a subsequent phase II trial. However, $d = 2$ is obviously more desirable than the MTD (ie, $d = 3$), with much lower toxicity and virtually identical efficacy. This example shows that for targeted therapy and immunotherapy, it is important to consider both toxicity and efficacy concurrently to identify the optimal biologic dose (OBD)⁴ rather than the MTD.

Various phase I/II dose-finding designs have been proposed to achieve this goal.⁵⁻¹⁵ Most of these designs assume statistical models to describe the dose toxicity and efficacy curves. As toxicity and efficacy data accrue during the trial, they continuously update the estimates of the models after each cohort and then use them to determine the dose for the next cohort (see Yuan et al⁶ for comprehensive reviews of phase I/II dose-finding designs). These novel model-based designs provide significant improvement over the conventional dose-finding methods in identifying the OBD. These designs are rarely used in practice, however,

ASSOCIATED CONTENT

Data Supplement

Author affiliations and support information (if applicable) appear at the end of this article.

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CONTEXT

Key Objective

For targeted therapies and immunotherapies, such as checkpoint inhibitors and chimeric antigen receptor T-cell therapy, where efficacy does not necessarily increase with the dose, the maximum tolerated dose may not be the optimal dose for treating patients. This article proposes a simple and efficient Bayesian adaptive phase I/II trial design, known as BOIN12, to find the optimal biologic dose (OBD) that optimizes the risk-benefit trade-off.

Knowledge Generated

BOIN12 is simple to implement: Its adaptation rule can be pretabulated and included in the protocol. During trial conduct, clinicians can simply look up the decision table to allocate patients to a dose without complicated computation. Compared with existing designs, BOIN12 has higher accuracy to identify the OBD and is safer.

Relevance

BOIN12 provides a simple, efficient, and robust design to optimize the dose and improve the success rate of targeted therapies and immunotherapies.

because they are often difficult to understand and require complicated statistical modeling and computation as well as an expensive infrastructure for implementation.¹⁶ We reviewed 13 phase I/II immunotherapy or targeted therapy trials¹⁷⁻²⁹ published from January 2018 to September 2019 (Data Supplement). All 13 trials used the conventional paradigm of finding the MTD first and then conducting CE.

The objective of this article is to remove these barriers and propose a transparent, easy-to-implement Bayesian optimal interval phase I/II (BOIN12) trial design to find the OBD. BOIN12 offers an extension of the toxicity-based BOIN design,^{30,31} which has been widely used in practice, to phase I/II dose-finding trials. Under BOIN12, patients are adaptively assigned to the most desirable dose with the optimal toxicity-efficacy trade-off. Implementation of BOIN12 does not require complicated computation, and the decision of dose escalation and de-escalation is easily made by looking up the decision table that can be pre-specified and included in the trial protocol.

METHODS

Toxicity-Efficacy Trade-off

BOIN12 uses a utility to measure the toxicity-efficacy trade-off, whose value should be elicited from clinicians to reflect the clinical desirability of each possible toxicity and efficacy outcome. For ease of exposition, we assume that toxicity and efficacy end points are binary, but the BOIN12 design can be generalized to multilevel or multiple outcomes (see the Data Supplement for an example of accounting for severe immune-related adverse events using a three-level toxicity end point). Given any patient in the trial, there are four possible outcomes: 1 = (no toxicity, efficacy); 2 = (no toxicity, no efficacy); 3 = (toxicity, efficacy); and 4 = (toxicity, no efficacy) (Table 1). (No toxicity, efficacy) is the most desirable, whereas (toxicity, no efficacy) is the least desirable, and the other two are in between. To elicit the utility for the four outcomes, we assign the most desirable outcome (no toxicity,

efficacy) a score of 100, and the least desirable outcome (toxicity, no efficacy) a score of 0 and then ask clinicians to use them as a reference to specify scores to the other two outcomes. Table 1 lists an example of elicited utility. In our experience, clinicians quickly understand what the utilities represent and provide their values. Several clinical trials using this approach have been implemented at The University of Texas MD Anderson Cancer Center.³²⁻³⁵ The purpose of assigning the scores of 0 and 100 to the least and most desirable outcomes is to fix the scale of utility, obtaining the commonly used 0-100 scoring system. The proposed design does not depend on the scale of the utility, and other scoring systems can also be used (eg, 1 and 10 to indicate the least and most desirable outcomes, respectively).

Let $u_1, \dots, u_4$ denote utilities ascribed to the four possible toxicity-efficacy outcomes, respectively. Given a specific dose d , let $p_1, \dots, p_4$ denote the respective probabilities of observing these outcomes at d , which typically vary across doses. Averaging over the four possible outcomes, the desirability (or mean utility) of d is

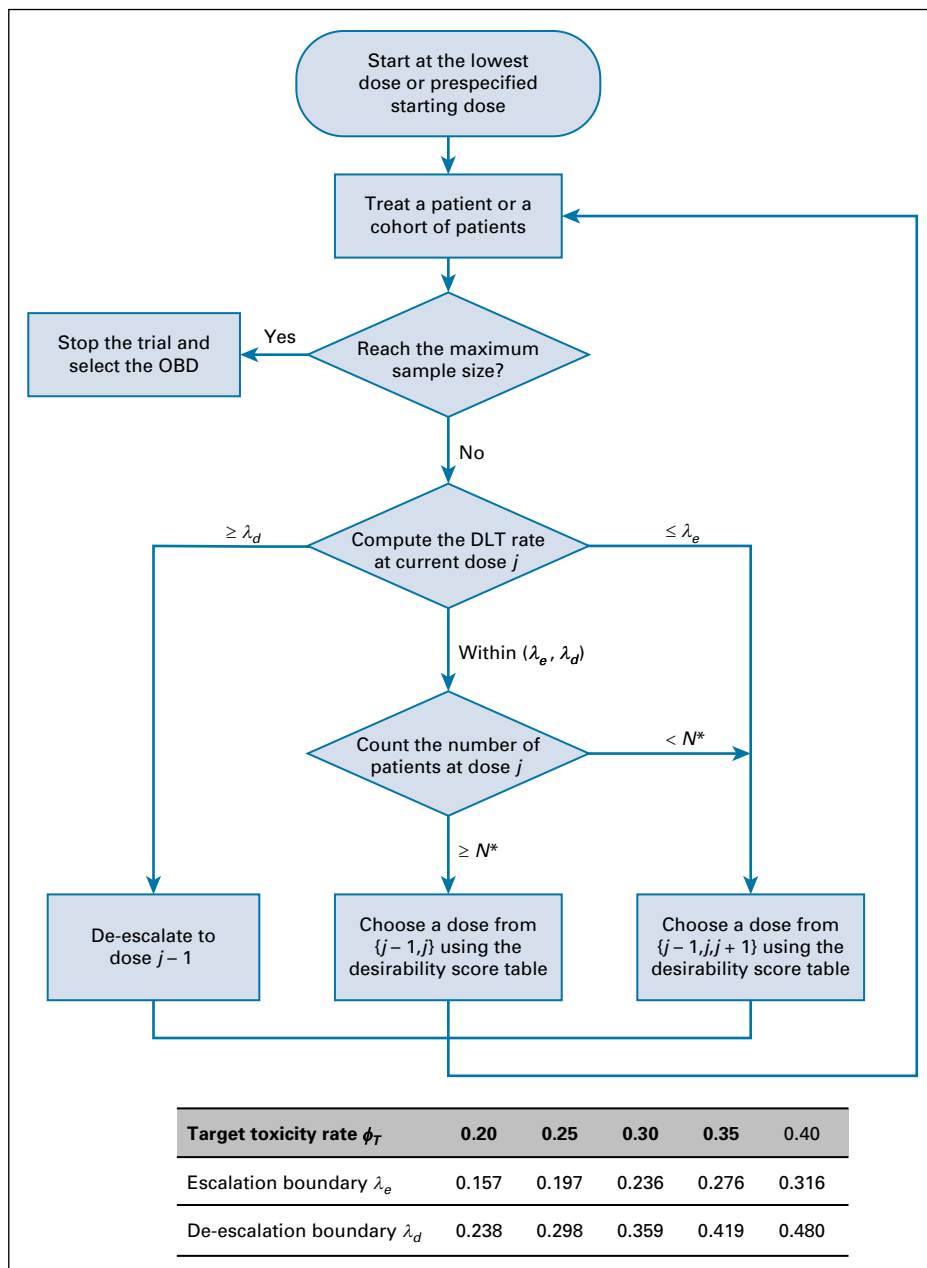
$$u(d) = p_1 u_1 + p_2 u_2 + p_3 u_3 + p_4 u_4 \quad (\text{Eq 1})$$

TABLE 1. Utility Table for Binary Efficacy and Toxicity End Points

Toxicity	Efficacy	
	Yes	No
No	$u_1 = 100$	$u_2 = 40$
Yes	$u_3 = 60$	$u_4 = 0$

NOTE. To fix the scoring scale, the most desirable outcome (no toxicity, efficacy) is assigned the highest utility score of $u_1 = 100$, and the least desirable outcome (toxicity, no efficacy) is assigned the lowest score of $u_4 = 0$. The scores ascribed to the other two outcomes (ie, u_2 , u_3) should be elicited from clinicians to reflect their clinical desirability: Setting $u_3 \geq u_2$ means that patients are willing to tolerate toxicity in exchange for efficacy; otherwise, $u_3 < u_2$ should be used. From the dose-finding perspective, setting $u_3 \geq u_2$ usually leads to a more thorough exploration of the dose space than setting $u_3 < u_2$.

FIG 1. The schema of the Bayesian optimal interval phase I/II (BOIN12) design, where (λ_e, λ_d) are a pair of optimized dose escalation and de-escalation boundaries adopted from the BOIN design, and N^* is a prespecified sample size cutoff (eg, $N^* = 6$). DLT, dose-limiting toxicity; OBD, optimal biologic dose.



A higher value of $u(d)$ indicates a higher desirability of dose d in the risk-benefit trade-off. The dose with the highest desirability is the OBD. Revisit the scenario where true toxicity probabilities $p_T = .05, .12, .27, .35$, and $.50$ and efficacy probabilities $p_E = .20, .35, .36, .37$, and $.38$ for five investigational doses. Assuming that the utility in Table 1 and the correlation between p_T and p_E is $.25$, the desirability for $d = 3$ is $u(d = 3) = 50.8$, and the desirability for $d = 2$ is $u(d = 2) = 56.2$. By using the utility approach, we naturally account for the risk-benefit trade-off and correctly identify that $d = 2$ is the OBD.

To safeguard patients from toxic and/or futile doses, two dose-acceptability criteria are used by BOIN12 to decide

which doses may be used to treat patients. Let ϕ_T and ϕ_E be the clinician-specified toxicity upper limit and efficacy lower limit, respectively. Generally, ϕ_E can take the value of the target response rate specified for a standard phase II trial. Because BOIN12 considers the toxicity-efficacy trade-off, the value of ϕ_T should be set slightly higher than the target toxicity rate used in conventional toxicity-based phase I designs. For example, if 30% is an appropriate target toxicity rate that the conventional phase I design used, then $\phi_T = 35\%$ or 40% is a reasonable choice for BOIN12. Let $p_T(d)$ and $p_E(d)$ denote the toxicity and efficacy probabilities of dose d , respectively. Dose d is defined as admissible if the observed data indicate that d is reasonably safe (eg, $\Pr[p_T(d) > \phi_T \mid \text{data}] < 0.95$) and efficacious (eg,

$\Pr p_E(d) > \phi_E | \text{data}] < 0.9$). During the trial, only admissible doses can be assigned to incoming patients, and doses that are not admissible should be eliminated. If all doses are eliminated, the trial is stopped without any dose selected.

Dose Escalation/De-escalation Rule

The dose-finding rule of the BOIN12 design is shown in Figure 1 and listed in Table 2. A key innovation of BOIN12 is that its calculation of dose desirability (in steps 2b and 2c in Table 2) can be pretabulated and included in the trial protocol before the trial starts, which greatly simplifies design implementation. This is achieved by using a novel statistical method, known as the quasi-beta-binomial model,³⁶ which converts complicated desirability calculations (that involve bivariate toxicity and efficacy outcomes) into simple beta-binomial modeling. Technical details are provided in the Data Supplement. Table 3 lists the rank-based desirability score (RDS) with a cohort size of three up to nine patients treated at a dose, with the highest acceptable toxicity probability $\phi_T = 0.35$ and the lowest acceptable efficacy probability $\phi_E = 0.25$. A larger value of RDS means a higher utility. Of note, the RDS is not the raw value of desirability, as defined in Equation 1, but the desirability rank among all possible outcomes, as explained in the Data Supplement. This allows for a simpler presentation of the desirability of a dose using integers.

During the trial, the desirability of a dose can be simply determined as follows: count the number of patients treated at that dose, the number of patients who experienced toxicity, the number of patients who experienced efficacy, and the number of patients who experienced efficacy

without toxicity (not needed when $u_2 + u_3 = 100$), then look up in Table 3 to determine its RDS. For example, suppose at a certain point of the trial, the numbers of patients treated at the first three doses are three, six, and three; the numbers of toxicities are zero, one, and two; and the numbers of efficacy outcomes are zero, three, and one. The current dose is $d = 2$. According to dose-finding rule listed in Table 2 (step 2c), because the observed toxicity rate $\hat{p}_T(d) = 0.167$ is less than the escalation boundary $\lambda_e = 0.276$, we look up in Table 3 and obtain that the RDS of each dose is 35, 56, and 31, respectively. Because dose level 2 has the highest RDS, the decision is to stay at the current dose for treating the next cohort of patients. Table 3 assumes a cohort size of three, but the BOIN12 design is flexible and can allow any pre-specified cohort size. The RDS table can be easily generated using the BOIN12 companion software. The Data Supplement lists another example of an RDS table with $u_2 + u_3 \neq 100$. The Data Supplement provides a step-by-step implementation of BOIN12 with application to the methotrexate plus erlotinib plus celecoxib trial.²⁸

For ethical consideration, BOIN12 always assigns the next cohort of patients to the dose with the highest estimate of desirability. With a small sample size, this myopic approach may cause the dose finding to get stuck at a locally optimal dose. To alleviate that issue, we impose a dose exploration rule: If the number of patients treated at the current dose d is greater than eight, the observed toxicity rate $\hat{p}_T(d)$ is less than the de-escalation boundary λ_d , and the next higher dose has never been used for treating patients, then we escalate the dose for treating the next cohort of patients.

Software

To facilitate the implementation of the BOIN12 design, we have developed a freely available web application.³⁷ The application consists of multiple functional modules, allowing users to generate the design flowchart (eg, Fig 1) and RDS table (eg, Table 3), to conduct simulation to obtain the operating characteristics of the design, and to generate protocol template for regulatory submission (see details in the Data Supplement).

Numerical Study: Simulation Configuration

We carried out simulation studies to evaluate the operating characteristics of the BOIN12 design. We considered a phase I/II dose-finding trial with five dose levels. The highest acceptable toxicity probability is $\phi_T = 0.35$, and the lowest acceptable efficacy probability is $\phi_E = 0.25$. The utility in Table 1 is used to define the toxicity-efficacy trade-off. Patients are treated in cohorts of three, with a maximum sample size of 36. We compared BOIN12 with the conventional design, the model-based efficacy-toxicity (EffTox) design,⁸ and the toxicity and efficacy probability interval (TEPI) design.¹³ The conventional design consists of a dose escalation phase using the 3 + 3 design followed by a CE at the identified MTD. Simon's two-stage design³⁸ is used to monitor efficacy in CE. We refer to this approach as the

TABLE 2. Decision Rule of the BOIN12 Design

1. Treat the first cohort of patients at the lowest or a prespecified starting dose.
2. Suppose at the current dose d , number n of patients who have been treated, among whom n_T had toxicity and n_E had efficacy. Define the observed toxicity rate as $\hat{p}_T(d) = n_T / n$, and let λ_e and λ_d denote the optimal dose escalation and de-escalation boundaries adopted from the BOIN design ^{30,31} (see Fig 1 for the values of λ_e and λ_d).
a. If $\hat{p}_T(d) \geq \lambda_d$, de-escalate the dose to $d - 1$ to treat the next cohort of patients.
b. If $\lambda_e < \hat{p}_T(d)$ and $n \geq 6$, determine the desirability of d and $d - 1$ using the prespecified RDS table (eg, Table 3), and choose the one that has the higher desirability to treat the next cohort.
c. Otherwise, ie, $\hat{p}_T(d) \leq \lambda_e$, or $\lambda_e < \hat{p}_T(d) < \lambda_d$ and $n < 6$, determine the desirability of $d - 1$, d , $d + 1$, using the prespecified RDS table (eg, Table 3) and choose the one that has the highest desirability to treat the next cohort.
3. Repeat step 2 until the maximum sample size is reached, and select the final OBD on the basis of the following two-step procedure (see the Data Supplement for estimation details):
a. Identify the MTD as the dose level that has the isotonicity estimated toxicity probability closest to the toxicity upper limit ϕ_T .
b. The final OBD is chosen as the dose level that has the highest estimated utility among the doses that are not higher than the MTD.

Abbreviations: BOIN12, Bayesian optimal interval phase I/II; MTD, maximum tolerated dose; OBD, optimal biologic dose; RDS, rank-based desirability score.

TABLE 3. RDS Table for the BOIN12 Design

No. Pts	No. Tox	No. Eff	Desirability Score	No. Pts	No. Tox	No. Eff	Desirability Score	No. Pts	No. Tox	No. Eff	Desirability Score
0	0	0	60	6	3	2	22	9	2	4	45
3	0	0	35	6	3	3	38	9	2	5	58
3	0	1	55	6	3	4	51	9	2	6	70
3	0	2	76	6	3	5	67	9	2	7	83
3	0	3	91	6	3	6	81	9	2	8	92
3	1	0	24	6	4	0	1	9	2	9	98
3	1	1	44	6	4	1	6	9	3	0	E
3	1	2	63	6	4	2	15	9	3	1	7
3	1	3	80	6	4	3	27	9	3	2	14
3	2	0	13	6	4	4	42	9	3	3	25
3	2	1	31	6	4	5	56	9	3	4	36
3	2	2	48	6	4	6	72	9	3	5	49
3	2	3	69	6	≥ 5	Any	E	9	3	6	61
3	3	Any	E	9	0	0	E	9	3	7	74
6	0	0	22	9	0	1	25	9	3	8	85
6	0	1	38	9	0	2	36	9	3	9	94
6	0	2	51	9	0	3	49	9	4	0	E
6	0	3	67	9	0	4	61	9	4	1	3
6	0	4	81	9	0	5	74	9	4	2	9
6	0	5	93	9	0	6	85	9	4	3	17
6	0	6	100	9	0	7	94	9	4	4	29
6	1	0	15	9	0	8	99	9	4	5	40
6	1	1	27	9	0	9	102	9	4	6	53
6	1	2	42	9	1	0	E	9	4	7	65
6	1	3	56	9	1	1	17	9	4	8	78
6	1	4	72	9	1	2	29	9	4	9	88
6	1	5	87	9	1	3	40	9	5	0	E
6	1	6	96	9	1	4	53	9	5	1	2
6	2	0	8	9	1	5	65	9	5	2	5
6	2	1	19	9	1	6	78	9	5	3	10
6	2	2	34	9	1	7	88	9	5	4	20
6	2	3	47	9	1	8	97	9	5	5	32
6	2	4	64	9	1	9	101	9	5	6	45
6	2	5	77	9	2	0	E	9	5	7	58
6	2	6	90	9	2	1	10	9	5	8	70
6	3	0	4	9	2	2	20	9	5	9	83
6	3	1	12	9	2	3	32	9	≥ 6	Any	E

NOTE. The upper toxicity limit $\phi_T = 0.35$, the lower efficacy limit $\phi_E = 0.25$, and the utility specifications are as in Table 1.

Abbreviations: BOIN12, Bayesian optimal interval phase I/II design; E, the dose should be eliminated because it does not satisfy the safety and efficacy admissible criteria (ie, not admissible because of high toxicity or low efficacy); Eff, efficacy; Pts, patients; RDS, rank-based desirability score; Tox, toxicity.

3 + 3 + CE design. The literature review described previously shows that nine (69.2%) of 13 published trials used this design. All the designs start the trial at the lowest dose level. The detailed configurations of competing designs and data generation procedure are provided in the Data Supplement.

We examined 16 representative scenarios with various dose-response curve shapes. Figure 2 shows nine representative scenarios, and the other scenarios are provided in the Data Supplement. We considered the following four performance metrics, on the basis of 10,000 simulated

trials, to quantify the operating characteristics of the designs: percentage of correct selection of the OBD, number of patients treated at the OBD, number of patients treated at the overdoses (ie, with toxicity rates > 0.35), and risk of poor allocation (ie, percentage of trials that allocate $< 36 / 5 = 7.2$ patients to the OBD).

RESULTS

The simulation results (see Fig 3) show that BOIN12 has the best overall performance among the four designs. It has the highest average percentage of correct selection of the OBD, and allocates the largest average number of patients to the OBD. In addition, BOIN12 is safer and more reliable than other designs, as demonstrated by the (nearly) smallest number of patients treated at toxic doses and the lowest risk of poor allocation. As expected, the 3 + 3 + CE design has the worst overall performance because that design ignores the toxicity-efficacy trade-off and does not target the OBD. The performance of the EffTox design

varies dramatically from one scenario to another (eg, scenario 4 v8), depending on how well the assumed model reflects the true dose-toxicity and -efficacy relationship. Compared with BOIN12, TEPI has uniformly poor accuracy to identify the OBD, higher (ie, more than doubled) likelihood of overdosing patients, and higher (ie, almost doubled) risk of poor allocation.

We also performed extensive sensitivity analyses to evaluate the performance of BOIN12 with a different sample size, starting dose level, correlation between toxicity and efficacy, and utility specification. Details are reported in the Data Supplement. The results suggest that BOIN12 is robust and generally outperforms other designs.

DISCUSSION

The proposed BOIN12 design provides a flexible framework to identify the OBD and account for risk-benefit trade-off underlying medical decisions in clinical trials. Thanks to

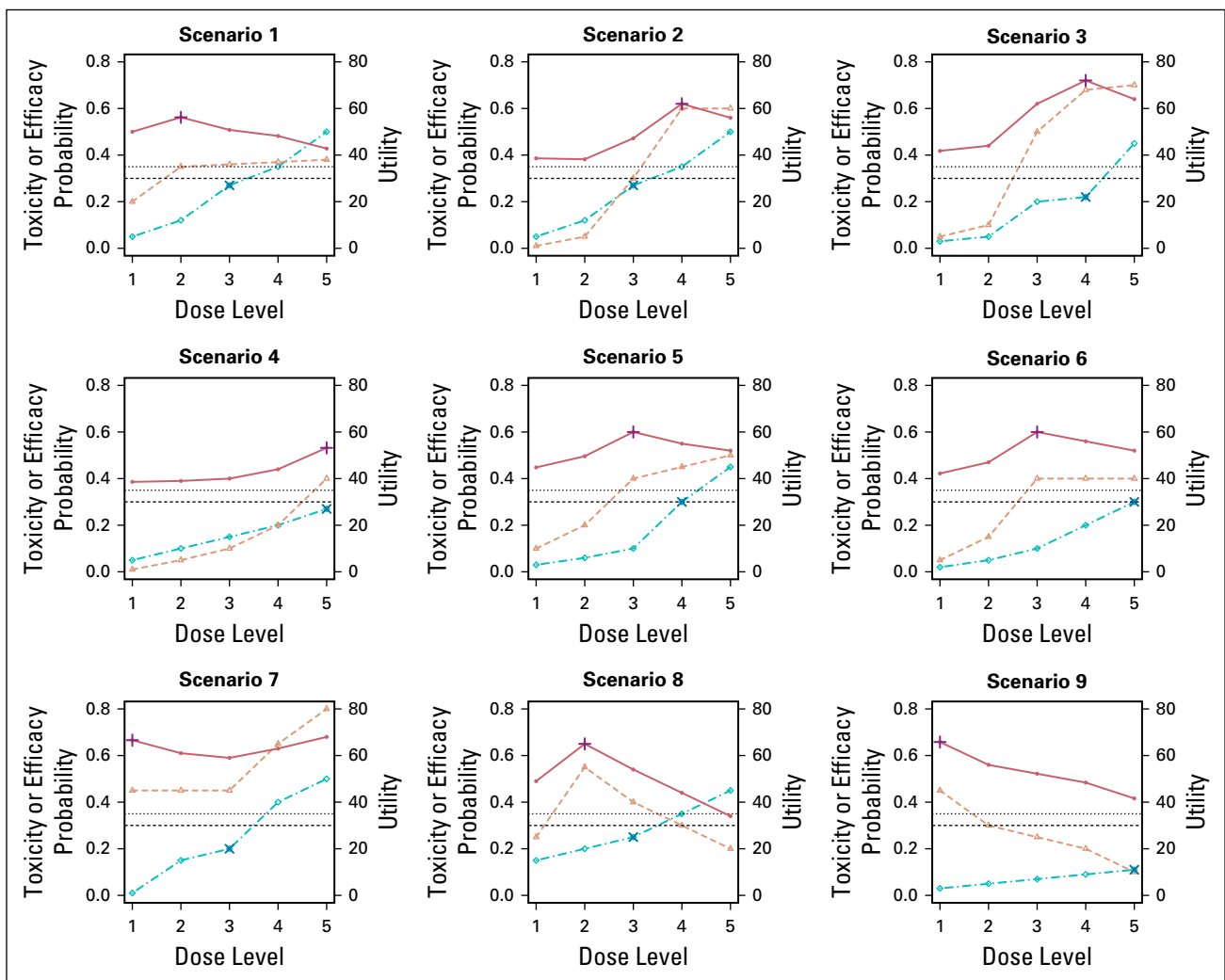


FIG 2. Nine dose-toxicity (dash-dotted line), dose-efficacy (gold dashed line), and dose-utility (solid line) scenarios considered in the simulation study. The horizontal dashed and dotted lines indicate the toxicity rate of 0.3 and 0.35, respectively. The minimum tolerated dose is marked by an x, and the optimal biologic dose, defined on the basis of the utilities listed in Table 1, is marked by a +.

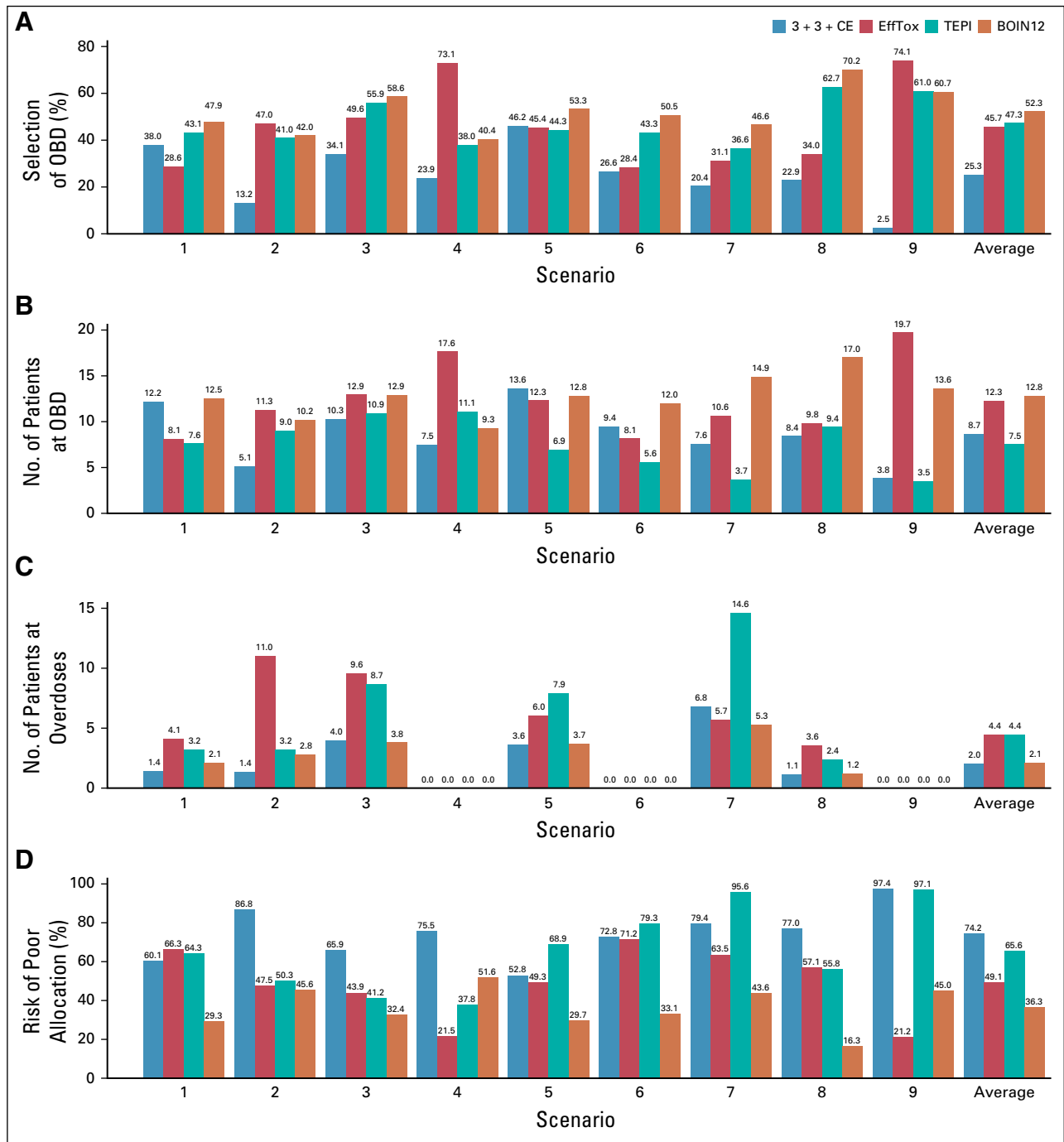


FIG 3. (A) Percentages of correct selection of the optimal biologic dose (OBD), (B) number of patients treated at the OBD, (C) number of patients treated at overdoses, and (D) risk of poor allocations for the 3 + 3 + cohort expansion (CE), efficacy-toxicity (EffTox), toxicity and efficacy probability interval (TEPI), and Bayesian optimal interval phase I/II (BOIN12) designs computed on the basis of 10,000 simulated trials under each of the nine scenarios shown in Figure 2. The overdoses are defined as the doses with toxicity rates > 0.35. The maximum sample size is 36. Risk of poor allocation is the percentage of trials that allocates < 36 / 5 = 7.2 patients to the OBD.

the use of the novel quasi-beta-binomial method, the BOIN12 design is simple to implement, and the decision of dose escalation and de-escalation can be easily made by looking up the RDS table prespecified in the protocol, thus removing one of the biggest impediments to using phase I/II dose-finding designs. The BOIN12 design is safer, has

higher accuracy to identify the OBD, and allocates more patients to the OBD. For ease of exposition, we focus on binary toxicity and efficacy outcomes; however, the BOIN12 design is also capable of handling multilevel toxicity and efficacy outcomes (eg, low, median, and high toxicity/efficacy) in a similar way (Data Supplement).

BOIN12 differs from existing dose-finding designs in several important ways. BOIN12 uses the dose escalation and de-escalation boundaries of the BOIN design; however, unlike the latter, which focuses on identifying the MTD using toxicity data only, BOIN12 accounts for the toxicity-efficacy trade-off by identifying the dose with the highest desirability for treating patients. Compared with the model-assisted phase I/II dose-finding designs,¹³⁻¹⁵ BOIN12 proactively and adaptively compares multiple doses to identify the optimal one for treating patients. Statistically, finding the OBD is to find the mode of the dose-desirability curve. To achieve that goal, it is necessary to examine multiple doses simultaneously (like rules 2b and 2c in Table 2). Making a dose escalation/de-escalation decision on the basis of only the data from the current dose, such as done with TEPI, is intrinsically insufficient and leads to the risk of missing the OBD, as shown in our numerical study. In addition, TEPI follows the modified toxicity probability interval design to use the unit probability mass to guide the decision of dose escalation/de-escalation decisions, thus inheriting the latter's drawback of a high risk of overdosing patients.^{39,40}

Compared with model-based phase I/II dose-finding designs, such as the EffTox design,⁸ which requires real-time complicated model fitting and estimation to make the decision of dose escalation/de-escalation, BOIN12 is much simpler to implement. In addition, BOIN12 is more robust than the EffTox design because it does not make any model assumptions on the dose toxicity and efficacy curves. BOIN12 differs from the recently proposed utility-based BOIN (U-BOIN) design in that U-BOIN is a two-stage design⁴¹ for which the first stage performs dose escalation on the basis of toxicity, as in conventional phase I trials,

and only the second stage uses the toxicity-efficacy trade-off for decision making. In contrast, BOIN12 is a single-stage design and uses the toxicity-efficacy trade-off throughout, thus being more efficient for finding the OBD.

Finally, compared with the existing designs, another important advantage of the BOIN12 design is its generalizability. In addition to the utility function in Equation 1, BOIN12 can be extended to accommodate various utility functions for quantifying the dose desirability. More discussions on incorporation of different utility functions can be found in the Data Supplement.

One limitation of the BOIN12 design is that it assumes that the toxicity and efficacy outcomes are observed quickly, such that the data of the previously treated patients are available by the time of next dose assignment. In some immunotherapy trials, toxicity and/or efficacy may be late onset,⁴² causing logistical difficulties when implementing the BOIN12 design. Under such circumstances, the missing data imputation method used in the time-to-event BOIN design⁴³ can be paired with the BOIN12 design to facilitate the real-time decision making of dose escalation/de-escalation. Another limitation of the BOIN12 design is that it does not consider patient heterogeneity (eg, patients are a mixture of sensitive and nonsensitive patients to immune checkpoint inhibitors⁴⁴). When the sensitive and nonsensitive subpopulations can be predefined (eg, by PD-L1 expression level), BOIN12 can be applied to each subgroup independently to identify subpopulation-specific OBD. However, when the patient subpopulations are unknown and need to be adaptively identified during the trial, a much larger sample size will be needed, and further research is warranted to generalize BOIN12 to handle this case.

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